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## Aideepen TFT Display Module with EC11 Rotary Encoder

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# User Manual: Aideepen TFT Display Module with EC11 Rotary Encoder

**Model:** TFT Display Module with EC11 Rotary Encoder (1.8-inch / 2.4-inch)

**Brand:** Aideepen

## 1. INTRODUCTION

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This module combines a TFT (Thin-Film Transistor) display screen with an EC11 rotary encoder, offering a compact and versatile human-machine interface solution. It is designed for various DIY electronic projects, particularly those involving microcontrollers like Arduino. The module is available in two display sizes: 1.8-inch and 2.4-inch, each featuring a high-resolution display and an integrated rotary encoder with a push-button switch.

### Key Features:

- **All-in-one design:** Display and encoder are integrated onto a single board for simplicity and aesthetic appeal, convenient for DIY projects.
- **Multiple size options:** Available in 1.8-inch and 2.4-inch TFT display sizes to meet diverse project requirements.
- **High-resolution display:** The 1.8-inch version offers 128x160 RGB resolution, while the 2.4-inch version provides 320x240 RGB resolution, ensuring clear and detailed visuals.
- **SPI interface support:** Utilizes a standard SPI interface for strong compatibility, making it easy to connect and program with various microcontrollers.
- **Encoder with switch:** Features an EC11 rotary encoder with a 15mm plum blossom shank, providing 20 pulses per rotation and 20 positioning points, along

with an integrated push-button switch function.

## 2. SAFETY INFORMATION

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Please read and understand the following safety precautions before using the module:

- Always ensure proper power supply voltage (3.3V/5V) to avoid damage to the module or connected components.
- Avoid touching the exposed circuit board components while the module is powered to prevent electric shock or electrostatic discharge.
- Handle the display screen with care to prevent scratches or breakage.
- Do not expose the module to extreme temperatures, humidity, or corrosive environments.
- Ensure all connections are secure and correctly wired before applying power.

## 3. PACKAGE CONTENTS

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Verify that all items are present and in good condition upon opening the package:

- TFT Display Module with EC11 Rotary Encoder x 1
- Pin headers x 1 (for connection)

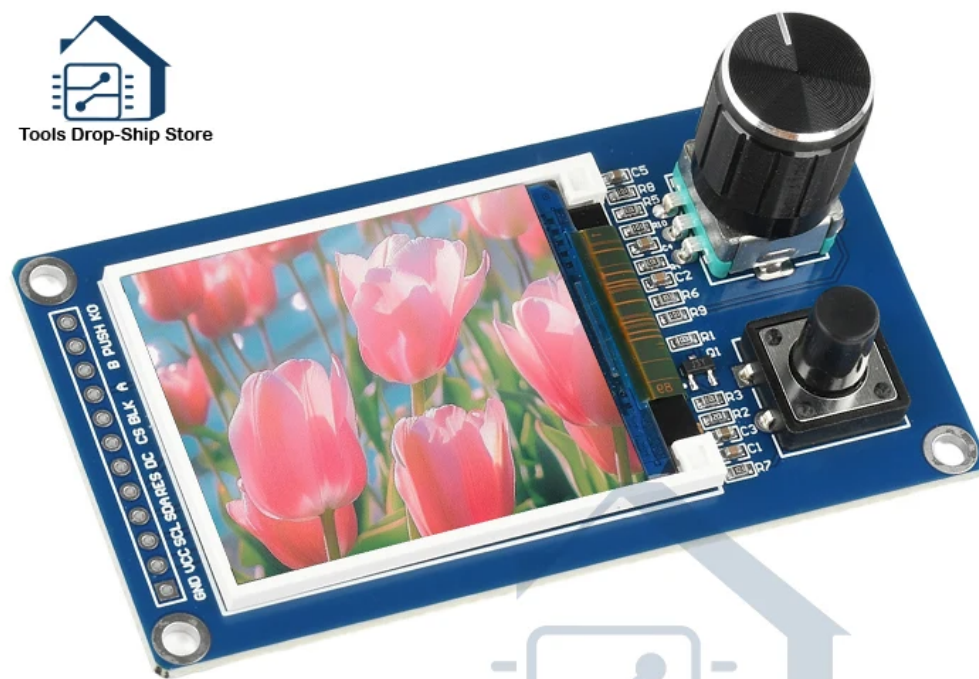


Figure 1: Overview of the TFT Display Modules with Rotary Encoders

## 4. SPECIFICATIONS

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### General Properties:

- **Type:** LCD display screen
- **DIY Supplies:** Electrical
- **Origin:** Mainland China
- **Certification:** CE

### Module Specific Parameters:

# 1.8' TFT+EC11:

4 fixing holes for M3 holes

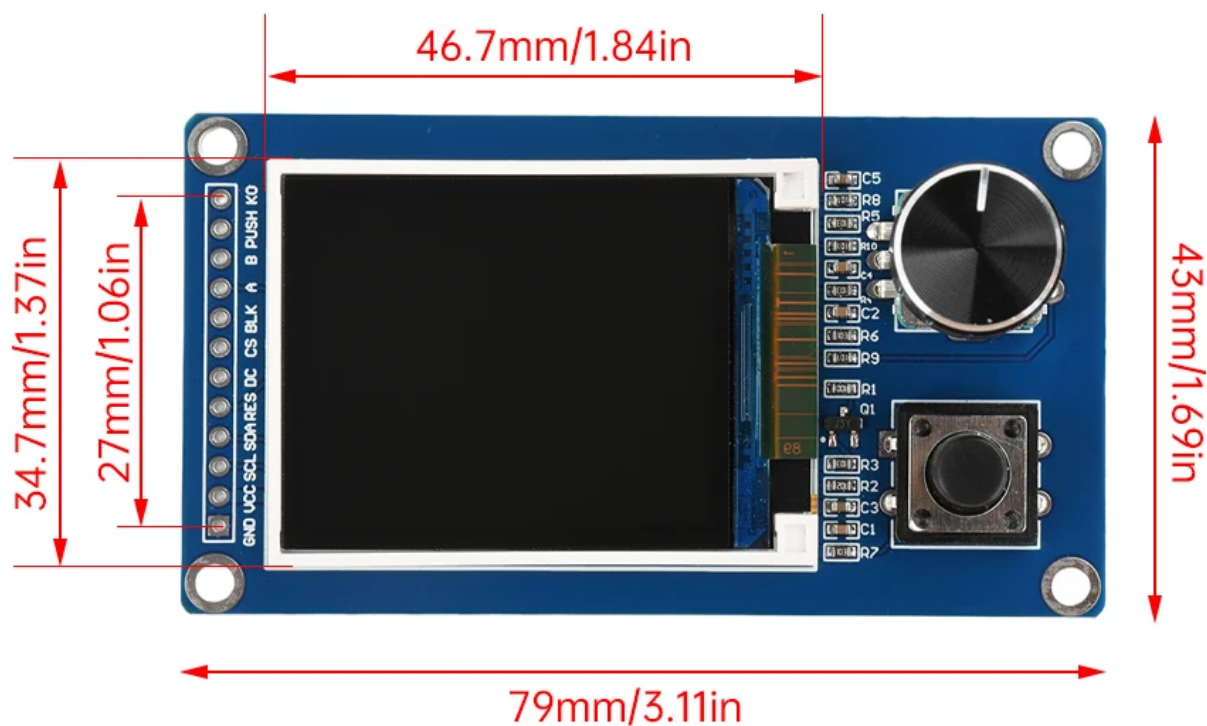


Figure 2: Product Parameter Comparison

**Table 1: Detailed Product Parameters**

| Property        | 1.8-inch TFT + EC11 Rotary Encoder                                                    | 2.4-inch TFT + EC11 Rotary Encoder                                                    |
|-----------------|---------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|
| Product Name    | 1.8' TFT + EC11 rotary encoder                                                        | 2.4' TFT + EC11 rotary encoder                                                        |
| TFT Driver Chip | ST7735S                                                                               | ST7789                                                                                |
| TFT Interface   | SPI                                                                                   | SPI                                                                                   |
| Resolution      | 128*160 RGB                                                                           | 320*240 RGB                                                                           |
| EC11            | Plum blossom shank, shank length 15mm, 20 pulses 20 positioning, 5 feet with a switch | Plum blossom handle, handle length 15mm, 20 pulses 20 positioning, 5 feet with switch |
| Touch Function  | No touch                                                                              | No touch                                                                              |

## Physical Dimensions:

## 2.4' TFT + EC11:

4 fixing holes for M3 holes

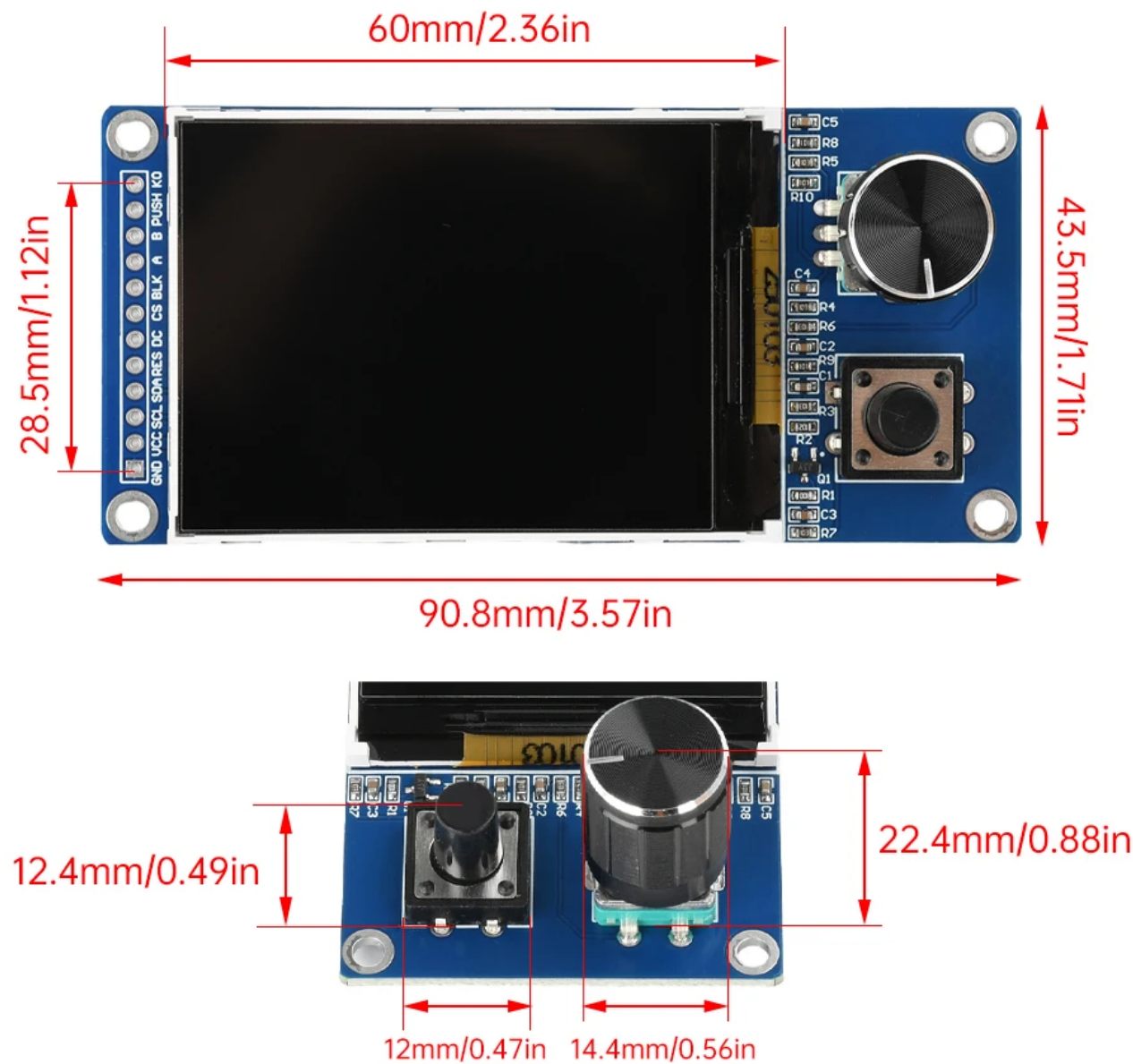


Figure 3: 2.4-inch Module Dimensions (e.g., 90.8mm x 43.5mm)

# Interface Description:

GND: Power source

VCC: 3.3V power supply positive

SCL: Clock interface for TFT SPI communication

SDA: Data interface for SPI communication on TFT screen

RES: Reset the pin

DC: Data command selection

CS: chip selection signal, low active

BLK:TFT backlight adjustment, fully enabled by default, you can not connect

K0: Can be used as a menu to return the key or confirm the key

PUSH: a key interface of the EC11

A, B: phases A and B of EC11

External Interface Pin

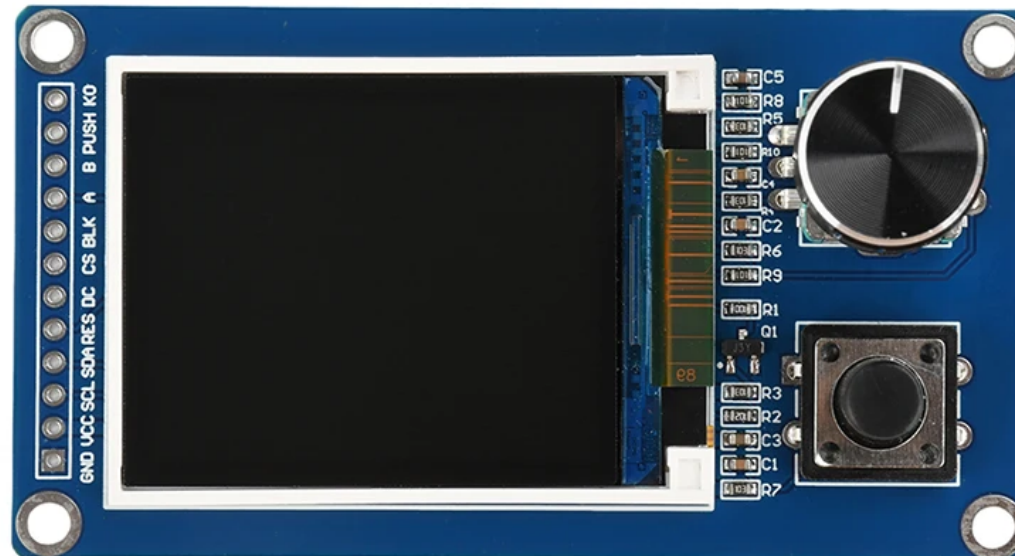
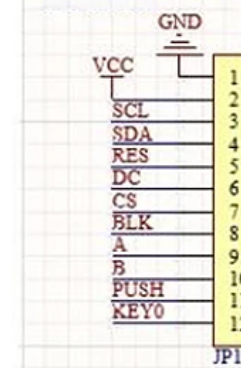


Figure 4: 1.8-inch Module Dimensions (e.g., 79mm x 43mm)

## 5. SETUP

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### 5.1 Interface Description

The module features a standard pin header for connection to your microcontroller. Refer to the pinout diagram and descriptions below for correct wiring:

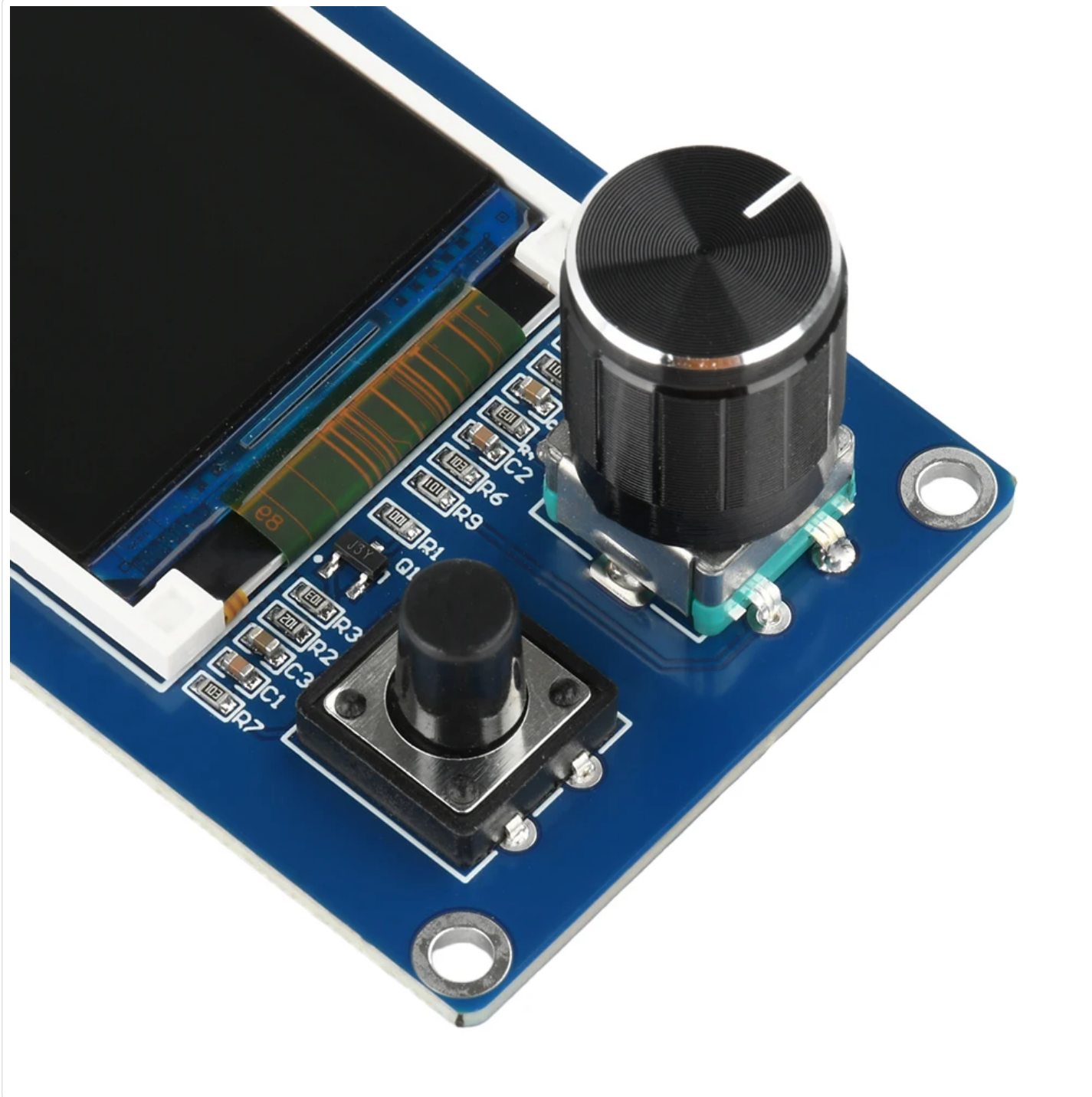


Figure 5: Module Interface Pinout

- **GND:** Ground connection (Power source)
- **VCC:** Power supply input (3.3V or 5V positive)
- **SCL:** Clock interface for TFT SPI communication
- **SDA:** Data interface for SPI communication on TFT screen
- **RES:** Reset pin for the display
- **DC:** Data/Command selection pin
- **CS:** Chip Select signal (low active)
- **BLK:** TFT backlight adjustment pin (fully enabled by default, can be left unconnected if not used for adjustment)
- **K0:** Additional key input, can be used as a menu return or confirmation key
- **PUSH:** Push-button switch input for the EC11 rotary encoder
- **A, B:** Phase A and Phase B outputs of the EC11 rotary encoder

## 5.2 Wiring Diagrams

The following diagrams illustrate the internal circuitry of the EC11 rotary encoder, TFT backlight, and the K0 confirmation key. These are provided for advanced users and troubleshooting; typical usage involves connecting the module's pins to a microcontroller's SPI and GPIO pins.

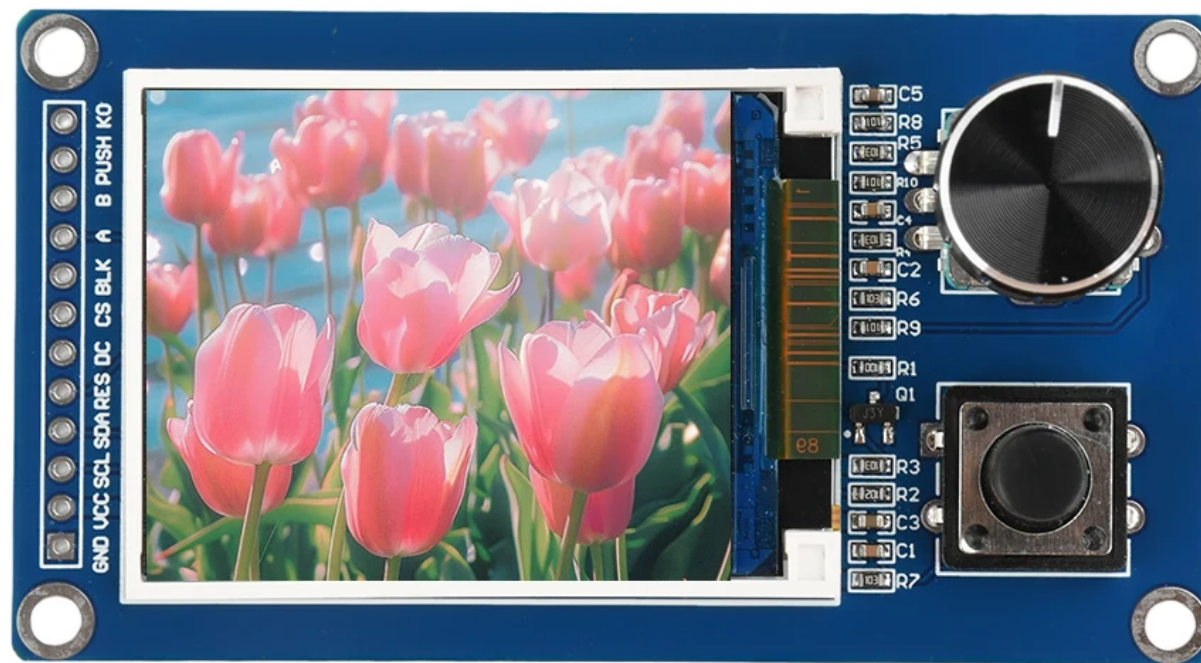


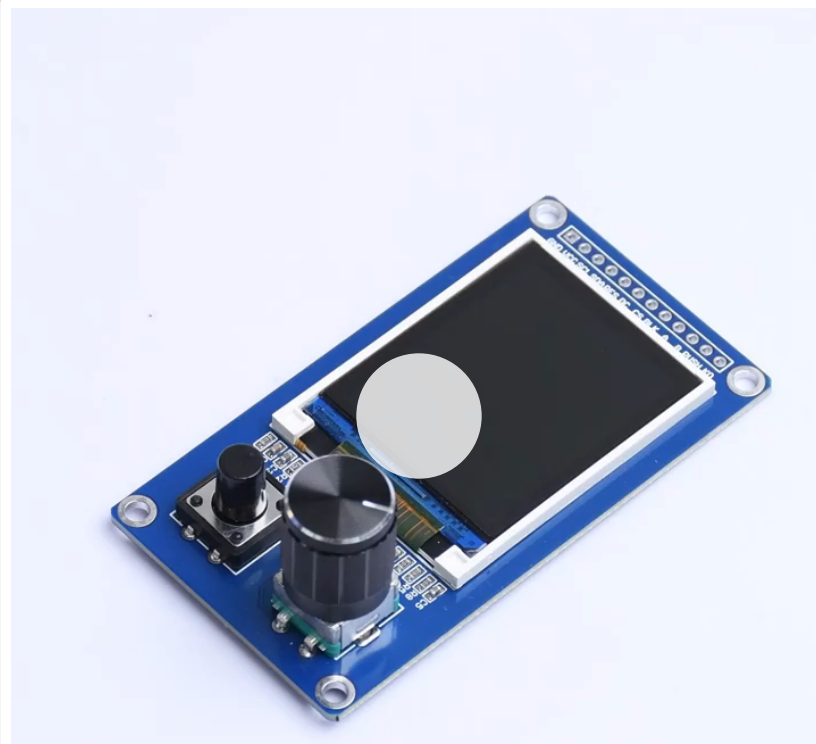
Figure 6: EC11 Rotary Encoder and TFT Backlight Schematics



Figure 7: K0 Confirmation Key Circuit Diagram

### 5.3 Initial Connection Steps (Example for Arduino)

1. **Power Supply:** Connect the module's VCC pin to your microcontroller's 3.3V or 5V output and GND to ground.
2. **SPI Connection:** Connect SCL, SDA, RES, DC, and CS pins to the corresponding SPI and GPIO pins on your microcontroller. Refer to your microcontroller's documentation for specific pin assignments.
3. **Encoder Connection:** Connect the A, B, and PUSH pins of the encoder to digital input pins on your microcontroller.
4. **K0 Key:** Connect the K0 pin to another digital input pin if you intend to use this additional key.
5. **Backlight (Optional):** The BLK pin can be connected to a PWM-capable pin on your microcontroller for backlight brightness control, or left unconnected for full brightness.



0:00 / 0:08

Video 1: Product Overview and Demonstration

## 6. OPERATING INSTRUCTIONS

Operating the TFT display and rotary encoder module involves programming your microcontroller to interact with its various components.

## 6.1 TFT Display Operation

The TFT display communicates via the SPI interface. You will need a suitable library for your microcontroller (e.g., Adafruit GFX and Adafruit ST7735/ST7789 libraries for Arduino) to initialize the display and draw graphics, text, or images. The resolution depends on the module size (128x160 for 1.8-inch, 320x240 for 2.4-inch).

## 6.2 Rotary Encoder Operation

The EC11 rotary encoder provides rotational input and a push-button switch. It has two phases (A and B) that output pulses when rotated, allowing your microcontroller to detect direction and amount of rotation. The PUSH pin acts as a standard momentary switch.

- **Rotation:** By monitoring the state changes on pins A and B, your microcontroller can determine if the encoder is being rotated clockwise or counter-clockwise. Each 'click' or 'detent' typically corresponds to a specific number of pulses (20 pulses per rotation for this model).
- **Push Button:** The PUSH pin will go low when the encoder shaft is pressed, and high when released. This can be used for selection, confirmation, or other menu interactions.

## 6.3 K0 Key Operation

The K0 key is an independent push-button switch. It can be programmed to perform various functions, such as returning to a previous menu, confirming an action, or toggling a setting. Like the encoder's push button, it typically provides a low signal when pressed.

## 7. USER TIPS

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- **Library Selection:** For Arduino, consider using the Adafruit GFX library along with the specific display driver library (Adafruit ST7735 for 1.8-inch, Adafruit ST7789 for 2.4-inch) for easy display control.
- **Encoder Debouncing:** Rotary encoders often require software debouncing to prevent false readings due to mechanical bounce. Implement a simple debouncing routine in your code.
- **Power Supply:** While the module supports both 3.3V and 5V logic, ensure your microcontroller's logic levels match the module's VCC to avoid potential damage or unreliable operation.
- **Code Examples:** Search online for example code for ST7735/ST7789 displays and EC11 rotary encoders with your chosen microcontroller platform (e.g., Arduino, ESP32) to get started quickly.

## 8. MAINTENANCE

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To ensure the longevity and optimal performance of your TFT Display Module with EC11

Rotary Encoder, follow these maintenance guidelines:

- **Cleaning:** Use a soft, dry, anti-static cloth to gently wipe the display screen and module surface. Avoid using abrasive cleaners, solvents, or excessive moisture.
- **Storage:** When not in use, store the module in a dry, dust-free environment, away from direct sunlight and extreme temperatures.
- **Handling:** Always handle the module by its edges to avoid touching the display surface or delicate electronic components.
- **Connections:** Periodically check all wiring connections to ensure they are secure and free from corrosion.

## 9. TROUBLESHOOTING

If you encounter issues with your module, try the following troubleshooting steps:

- **Blank Screen:**
  - Check power connections (VCC and GND).
  - Verify SPI wiring (SCL, SDA, RES, DC, CS) for correct pin assignments and secure connections.
  - Ensure the display is correctly initialized in your code with the appropriate driver (ST7735 for 1.8-inch, ST7789 for 2.4-inch).

- Confirm the BLK pin is either connected to a high signal or left floating (for full brightness).
- **No Encoder Response:**
  - Check connections for A, B, and PUSH pins to your microcontroller's digital inputs.
  - Ensure your code is correctly reading the encoder pins and implementing debouncing.
  - Test the encoder mechanically to ensure it clicks and rotates freely.
- **Incorrect Display Output:**
  - Double-check the display resolution and orientation settings in your code.
  - Verify that the correct display driver library is being used for your specific module size.
  - Ensure your microcontroller's SPI clock speed is compatible with the display module.
- **Module Not Powering On:**
  - Confirm the power supply voltage is within the specified range (3.3V-5V).
  - Check for any short circuits on the module or wiring.

## 10. WARRANTY AND SUPPORT

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This product is manufactured to high-quality standards. For technical support or inquiries, please contact your retailer or the manufacturer directly. Please retain your proof of purchase for any warranty claims.

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Model number, symptoms, what you tried, or the section of the manual you are using.

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